

First Year Master of Technology**Lab Assignment 4****Course Code:** 2304514L**Course Name:** Machine Learning Laboratory**Academic Year:** 2025–26

Student Details

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Title: Clustering Analysis on Mall Customer Dataset**Objective**

To implement clustering algorithms such as K-Means and Hierarchical Clustering on a real-world dataset in order to identify meaningful patterns and group similar data points, and to evaluate the quality of clustering using metrics like Silhouette Score and Davies-Bouldin Index.

Problem Statement

1. Implement clustering algorithms on a real-world dataset to identify meaningful groups within the data.
2. Evaluate the quality of clustering results using appropriate metrics such as silhouette score, Davies Bouldin index, or visual inspection.

Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score, davies_bouldin_score

# ----- LOAD DATA -----
df = pd.read_csv('/content/Mall_Customers.csv')

# Select features
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]

# Scale data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```

# ----- ELBOW METHOD -----
wcss = []

for i in range(2, 10):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(2,10), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()

# ----- K-MEANS -----
kmeans = KMeans(n_clusters=5, random_state=42)
k_labels = kmeans.fit_predict(X_scaled)

# ----- HIERARCHICAL -----
agg = AgglomerativeClustering(n_clusters=5)
a_labels = agg.fit_predict(X_scaled)

# ----- EVALUATION -----
def evaluate(X, labels, name):
    sil = silhouette_score(X, labels)
    db = davies_bouldin_score(X, labels)
    print(f"\n{name}")
    print("Silhouette Score:", sil)
    print("Davies-Bouldin Index:", db)

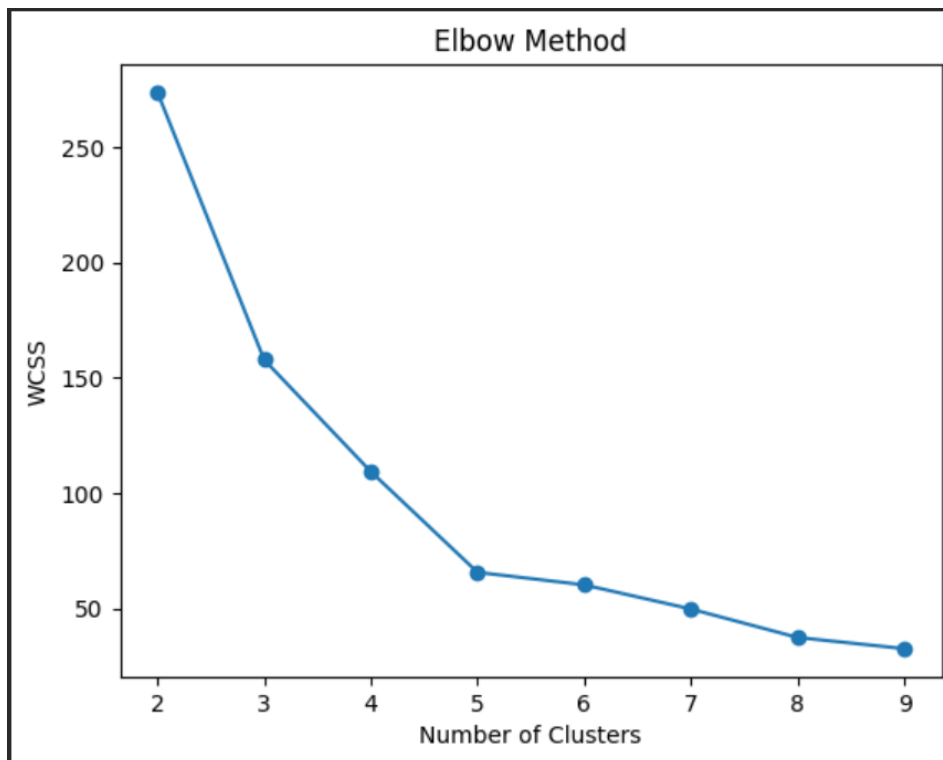
evaluate(X_scaled, k_labels, "K-Means")
evaluate(X_scaled, a_labels, "Hierarchical Clustering")

# ----- VISUALIZATION -----
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=k_labels, cmap='viridis')
plt.title("K-Means Clustering")
plt.xlabel("Income")
plt.ylabel("Spending Score")
plt.show()

plt.scatter(X_scaled[:,0], X_scaled[:,1], c=a_labels, cmap='rainbow')
plt.title("Hierarchical Clustering")
plt.xlabel("Income")
plt.ylabel("Spending Score")
plt.show()

```

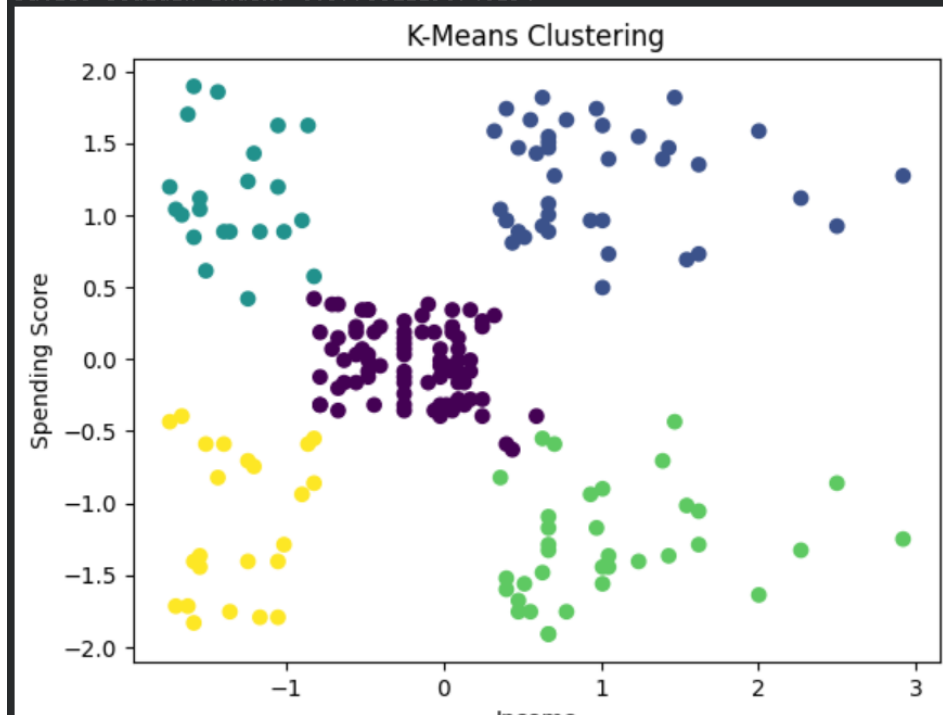
Output



```
K-Means  
Silhouette Score: 0.5546571631111091  
Davies-Bouldin Index: 0.5722356162263352
```

The Elbow Method indicates that the optimal number of clusters is 5, as the rate of decrease in WCSS significantly slows down after this point.

```
Hierarchical Clustering  
Silhouette Score: 0.5538089226688662  
Davies-Bouldin Index: 0.5778621190740154
```

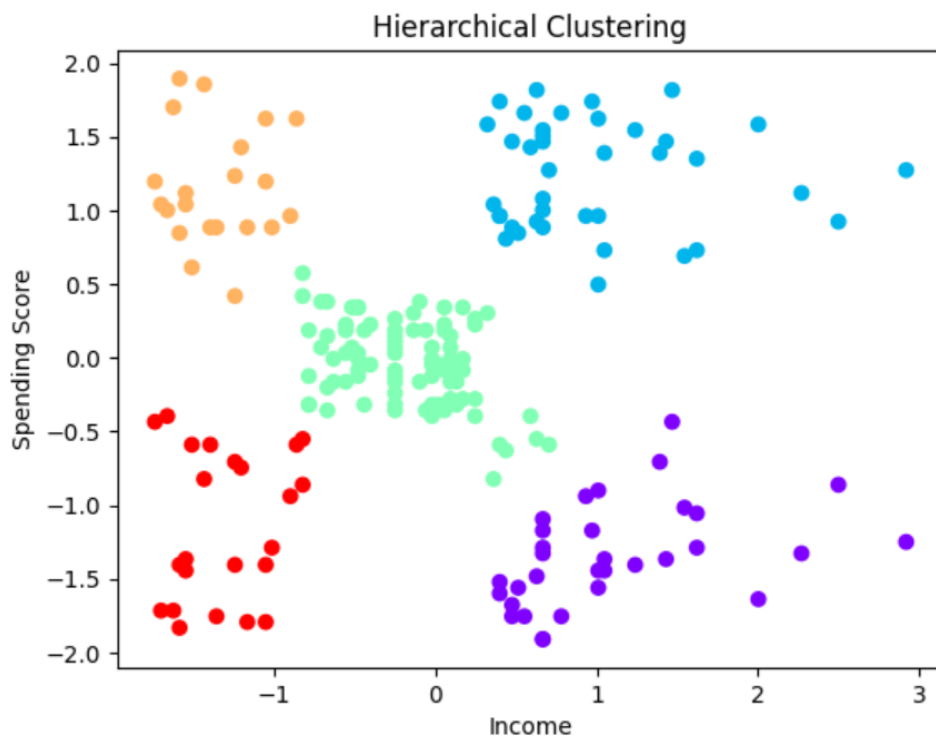


Algorithm	Silhouette Score	Davies-Bouldin Index
K-Means	0.5546	0.5722
Hierarchical	0.5538	0.5779

Values above 0.5 indicate reasonably good cluster separation

K-Means produces slightly more compact and well-separated clusters.

K-Means forms well-defined clusters with minimal overlap.



Hierarchical clustering captures structure but is slightly less precise.

Factor	K-Means	Hierarchical
Speed	Fast	Slower
Cluster Quality	Better	Slightly lower
Interpretability	Moderate	High
Scalability	High	Low

Writes on following points

1. What is the difference between supervised and unsupervised learning?

Supervised learning uses labeled data, where the output is already known. The model learns a mapping from input features to target labels. Examples include classification and regression.

Unsupervised learning uses unlabeled data, where no target output is given. The model tries to discover hidden patterns or structures in the data. Clustering is a common example.

So, the main difference is that supervised learning predicts known outputs, while unsupervised learning finds patterns without predefined labels.

Aspect	Supervised Learning	Unsupervised Learning
Data	Labeled	Unlabeled
Goal	Predict output	Find hidden patterns
Example	Classification, Regression	Clustering, Association

2. How does k-means clustering work, and what role does the value of k play?

Working:

1. Choose number of clusters (**k**)
2. Initialize centroids
3. Assign points to nearest centroid
4. Update centroids
5. Repeat until convergence

Role of k:

- Small k → Underfitting (large clusters)
- Large k → Overfitting (too many small clusters)

Choosing k correctly is critical (Elbow / Silhouette method).

3. Advantages and limitations of hierarchical clustering vs k-means

Advantages:

- No need to predefine k
- Produces **dendrogram (tree structure)**
- Works well for smaller datasets

Limitations:

- Computationally expensive
- Not scalable
- Sensitive to noise

Compared to K-Means:

- K-Means → Fast, scalable
- Hierarchical → More informative but slower

4. Explain the silhouette score

The silhouette score measures how well each point fits within its assigned cluster.

It is based on:

- **a** = average distance between a point and other points in the same cluster
- **b** = average distance between a point and points in the nearest others cluster

Formula:

$$S = \frac{b - a}{\max(a, b)}$$

Interpretation:

- **Close to 1** → point is well-clustered
- **Close to 0** → point lies between clusters
- **Negative value** → point may be assigned to the wrong cluster

It helps evaluate clustering quality by showing whether clusters are compact and well-separated.

5. Why is feature scaling important in clustering algorithms?

Feature scaling is important because clustering algorithms often rely on distance calculations such as Euclidean distance.

If one feature has a much larger range than another:

- it can dominate the distance calculation
- the clustering result may become biased toward that feature

For example:

- Annual Income may range in tens

- Spending Score may range from 1 to 100

Without scaling, the algorithm may give unequal importance to features. Scaling ensures all features contribute fairly.

Conclusion

The clustering algorithms were successfully implemented on the Mall Customer dataset to identify meaningful groups. The Elbow Method suggested an optimal number of clusters as 5. Both K-Means and Hierarchical clustering produced reasonably good results, as indicated by silhouette scores around 0.55.

K-Means slightly outperformed Hierarchical clustering, achieving a higher silhouette score and lower Davies-Bouldin index, indicating better cluster compactness and separation. Visual inspection further confirmed that K-Means produced more distinct and well-separated clusters.

Thus, K-Means is more suitable for this dataset due to its efficiency and ability to form compact clusters, while Hierarchical clustering provides better interpretability but at the cost of slightly reduced performance.

“Clustering performance depends on data distribution; K-Means works best for spherical clusters, while hierarchical methods are useful for understanding data structure.”